

EEGNet

Open Data & Analytics for Global Neuroscience



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EEGNet is an **open-source hub** for standardised electroencephalography (EEG) research — **bringing** together data, workflows, and computational infrastructure to advance brain health discovery worldwide. A pan-Canadian initiative supported by Brain Canada, and a core component of both the Canadian Open Neuroscience Platform ([CONP](#)) and the Global Brain Consortium ([GBC](#)).

13+

OPEN DATASETS

1500+

EEG RECORDINGS

300+

PLATFORM USERS

30+

COUNTRIES

THE CHALLENGE

Democratizing Brain Research Through Open Infrastructure

Most of what we understand about the brain derives from a **narrow, unrepresentative slice** of the global population. Large-scale neuroimaging has been dominated by MRI — costly, facility-dependent, and inaccessible across much of the world.

EEG changes this equation. Portable, affordable, non-invasive, and uniquely suited to vulnerable, paediatric, and ageing populations, EEG offers the temporal resolution needed to capture the dynamics of brain function — and the **scalability** to do it globally.

Yet the field has been held back by fragmented data formats, inconsistent metadata, limited sharing infrastructure, and the bottleneck of expert-driven manual annotation. EEGNet was built to remove these barriers leveraging a global network of neuroelectrophysiology research and clinical experts.

STRATEGIC ALIGNMENT

EEGNet provides an open interactive platform for consortia including [#EEGManyLabs](#) and is a core component of [EEG101](#), embedding **FAIR** data principles, **open science** and internationally adopted **data standards** at every layer of design.

COMMUNITY-DRIVEN PRIORITIES

A recent survey of 500+ EEG researchers confirmed: **barriers to data sharing** and the absence of a **standards-compliant analytics hub** are among the field's most pressing technical concerns.

([Mushtaq et al. 2024 Nat.Hum.Behav](#))



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What Makes EEGNet Distinctive



First in Visualization and embedding BIDS + HED at Scale

EEGNet is the first platform to integrate BIDS + HED with interactive annotation - enabling machine-readable, collaborative knowledge-building on shared data.

Global Health Equity

Linked to GBC population screening programmes in Low- and Middle-Income Countries, EEGNet enables research that goes beyond WEIRD population samples.

End-to-End Research Lifecycle

From acquisition and standardisation through secure storage, collaborative annotation, querying, and HPC analytics - all in a single interoperable ecosystem.



EEGNet Demonstration video

Request an Account and Explore at EEGNet.LORIS.ca



01 · DATA PLATFORM

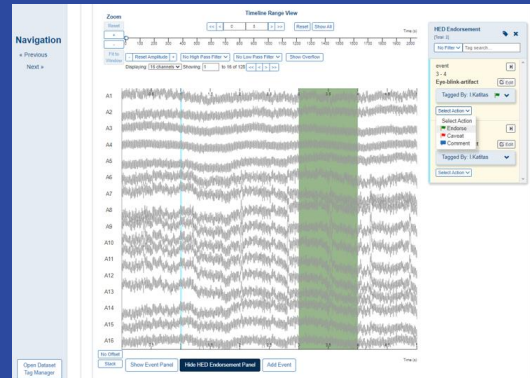
Visualization in-browser



Built on LORIS, EEGNet displays EEG recordings in a browser, including raw and derived BIDS-format datasets, with annotation layers. It is fully scalable for large-scale recording, discoverable via the CONP Portal and governed by CONP-G4GH ethics policies.

02 · OPEN WORKFLOWS

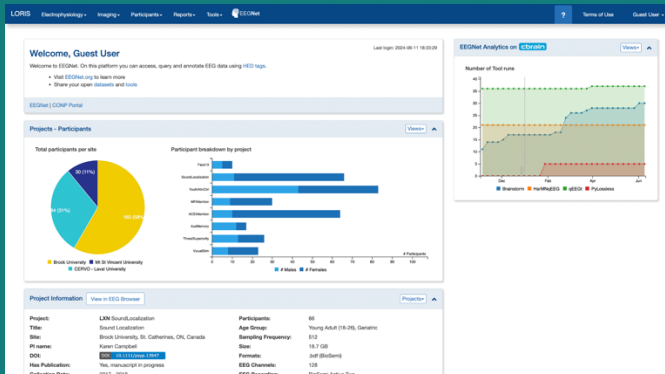
Interactive HED Annotation & Visualization



The Electrophysiology Browser enables real-time EEG signal visualization, with collaborative workflows for HED-SCORE clinical annotation, event flagging, and endorsement - features unique to LORIS and EEGNet among open repositories and software platforms worldwide.

03 · DATA SHARING

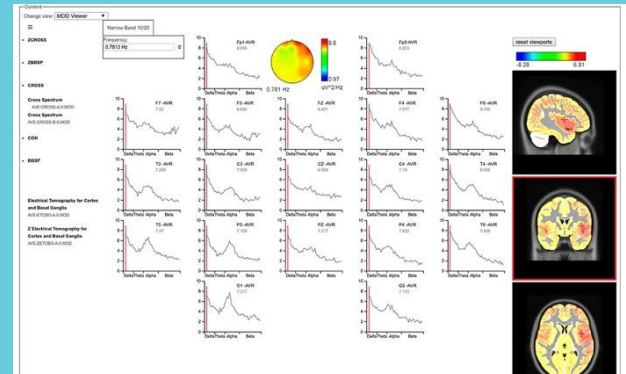
Custom Views and Querying



Open research assets, including 5 pipelines, 13+ open datasets, e.g. #EEGManyLabs resting data, structural MRI, standardized metadata, traceable data provenance and versioned releases.

04 · ANALYTICS HUB

Open Analytics: Big Computing on CBRAIN



Integration with CBRAIN provides containerized, reproducible EEG pipelines, e.g. HarMNqEEG, PyLossless, Brainstorm, and qEEG, deployed on the CBRAIN high-performance computing portal for global access.

Infrastructure Built on Proven Foundations



INFRASTRUCTURE

LORIS + CBRAIN

LORIS is a multimodal database used for e.g. ABCD, HBCD, UK Biobank, BigBrain. CBRAIN connects HPC clusters and cloud for large-scale data processing.

STANDARDS

BIDS & HED

All data are compliant with EEG-BIDS. HED tags and HED-SCORE clinical ontologies are modifiable on EEGNet.

TOOLS

EEGStudyFlow

Open-source dynamic notebooks ([EEGStudyFlow](#)) and cross-platform converter ([eeq2bids](#)) enable EEG-BIDS conversion on any operating system.

PIPELINES

HarMNqEEG & PyLossless

GBC's HarMNqEEG toolkit establishes quantitative EEG global norms. Brainstorm and PyLossless (ICA-based preprocessing tool) are also supported on CBRAIN.

GOVERNANCE

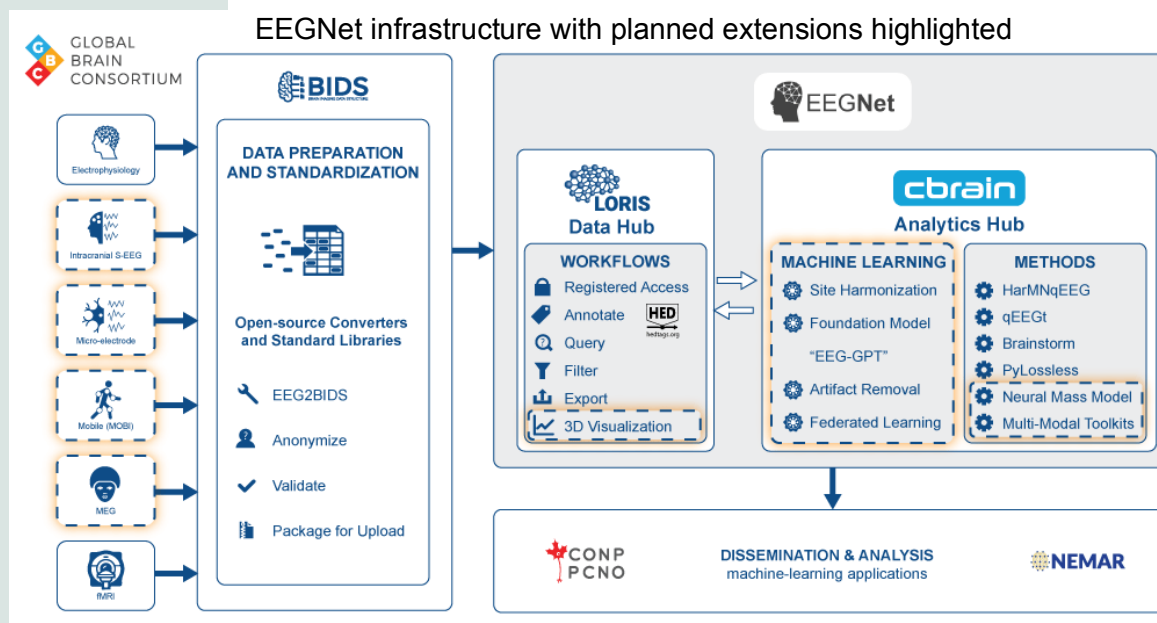
CONP Data Governance

EEGNet operates under Terms of Use from the CONP Ethics and Governance working group, aligned with G4GH and emerging international consensus models.

PUBLISHING

Evidence & EEGStudyFlow

Interactive Jupyter notebooks walk researchers through complete EEG analyses, published on the [Evidence](#) open publishing platform for full reproducibility.



EEG for the World's Population

Most brain datasets originate from Western, Educated, Industrial, Rich and Democratic (WEIRD) populations - despite the global majority and highest burden of neurological conditions residing in Low- and Middle-Income Countries (LMICs).

The Global Brain Consortium (GBC) is an international network spanning 30+ countries that uses an open data infrastructure for brain health research and translation for global health.

EEGNet has been part of the GBC's Global EEG Norms project since 2020, serving as the data hub for investigations into brain disorders in underserved populations - particularly in early childhood and adolescence.

EEG's portability and affordability make EEGNet uniquely positioned for data-driven collaborations and diagnostic tool development at scale for global health research.

GBC Member Countries	30+
Open Datasets	13
EEG Recordings	1,500+
MRI Records	141
Analytics Tools (CBRAIN)	5
Platform Users	300+
Published Outputs	25+

The EEGNet paradigm shift - From isolated data silos and inconsistent standards to global open sharing of tools and data. Spanning the research lifecycle from acquisition to publication.

STRATEGIC ROADMAP

Future Directions

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i Extending Datasets, Formats & Workflows

Additional data formats and expanded dataset collections are anticipated as the platform scales. Future workflow development will include granular collaborative annotation and validation controls, advanced metadata querying, and enhanced visualisation — guided by the GBC and EEGNet's pan-Canadian research network.

ii Machine Learning for Biomarker Identification

Integration of machine learning tools optimized for standardized EEG data is a primary future EEGNet capability. Translational applications will target earlier detection developmental disorders and neurodegenerative disease.

iii Automation of Data Quality & Upload

Automated data upload via LORIS's EEG Uploader module, validated signal quality metrics from the PyLossless pipeline, and automated quality assessment tools for contributed datasets are all in active development.

iv Global Accessibility & Standards Acceleration

The GBC, in partnership with EEGNet, is working to lower the technical and financial costs of participation in open neuroscience research across the world, particularly for institutions in resource-constrained settings.

Publications and Presentations | EEGNet.org/publications.html

Contact & Resources

Platform

EEGNet.Loris.ca

EEGStudyFlow

Infrastructure

LORIS Data Platform

CBRAIN Analytics Portal

CONP Portal

Contact & Support

EEGNet.org

contact@eegnet.org EEGNet

videos on Youtube

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Key Standards

[BIDS \(Brain Imaging Data Structure\)](https://BIDS (Brain Imaging Data Structure))

[HED \(Hierarchical Event Descriptors\)](https://HED (Hierarchical Event Descriptors))

Core Partners

Global Brain Consortium

GBC on YouTube

CONP.ca

#EEGManyLabs | EEG101

